

 **TARGET CAT 2025** 

MBA FASTRACK

QUANT : Arithmetic

Mixture & Allegation

Lecture No.- 05

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Recap of Previous Lecture :

$$\boxed{\begin{array}{r} CP \\ + P \\ \hline SP \end{array}} \quad \boxed{\begin{array}{r} CP \\ - L \\ \hline SP \end{array}} \quad \boxed{\begin{array}{r} MP \\ - Disc \\ \hline SP \end{array}}$$

P.J. or L.J. $\rightarrow$ CP

Disc ./. $\rightarrow$ MP

$$SP = (100 + P) \cdot / \cdot CP$$

$$SP = (100 - L) \cdot / \cdot CP$$

$$SP = (100 - Disc) \cdot / \cdot MP$$



Recap of Previous Lecture :

$$P.\% = \frac{\text{Profit}}{C_p} \times 100$$

$$L.\% = \frac{\text{Loss}}{C_p} \times 100$$

$$\text{Disc.}\% = \frac{\text{Disc}}{M_p} \times 100$$

TOPICS *to be covered*



- 1) Concept & Numerical based on 'Mixture & Allegation'
- 2) Concept & Numerical based on 'Replacement & Removal'
- 3) Problems for Practice & CAT PYQs

QUESTION-1

$$120 - 12 = 108$$

$$\begin{array}{r} 10\% \rightarrow 350 \\ 5\% \rightarrow 175 \\ \hline 525 \end{array}$$



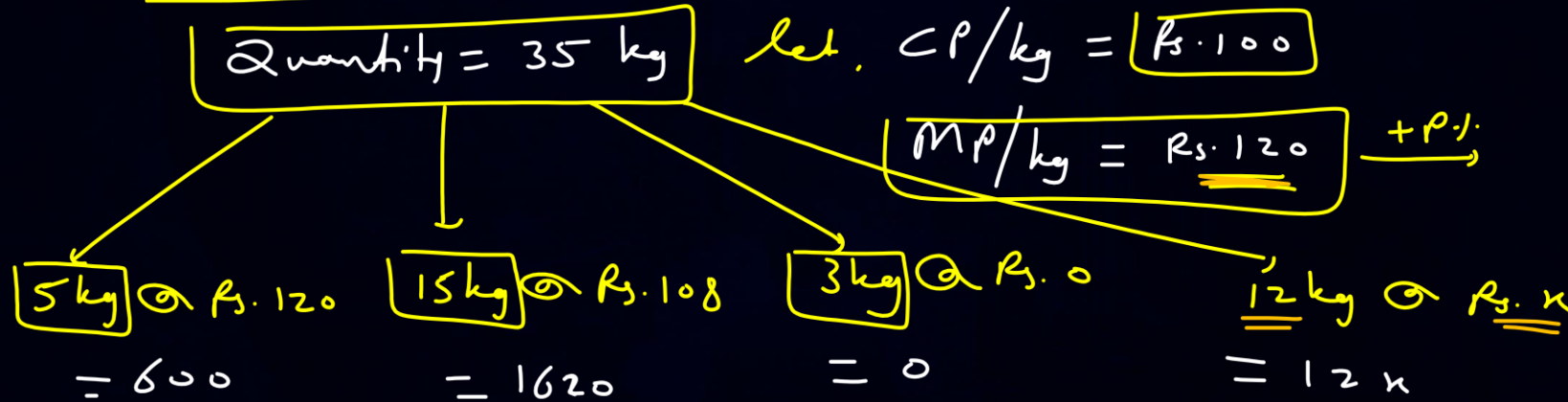
#Q. A man buys 35 kg of sugar and sets a marked price in order to make a 20% profit. He sells 5 kg at this price, and 15 kg at a 10% discount. Accidentally, 3 kg of sugar is wasted. He sells the remaining sugar by raising the marked price by p percent so as to make an overall profit of 15%. Then p is nearest to [CAT 2020 : Slot 3]

A 22

B 35

C 31

D 25 ✓



$$\begin{array}{l} \text{T. Cost} \\ = (35 \times 100) \\ = 3500 \\ + \\ (15\%) \rightarrow 525 \\ \hline \text{Sale Target } 4025 \end{array}$$

$$600 + 1620 + 12x = 4025$$

$$12x = 4025 - 2220$$

$$12x = 1805$$

$$x = 150 \dots$$

$$120 \xrightarrow{+30} 150$$

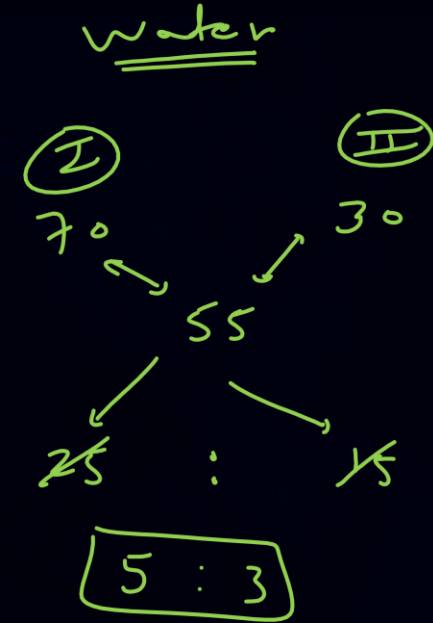
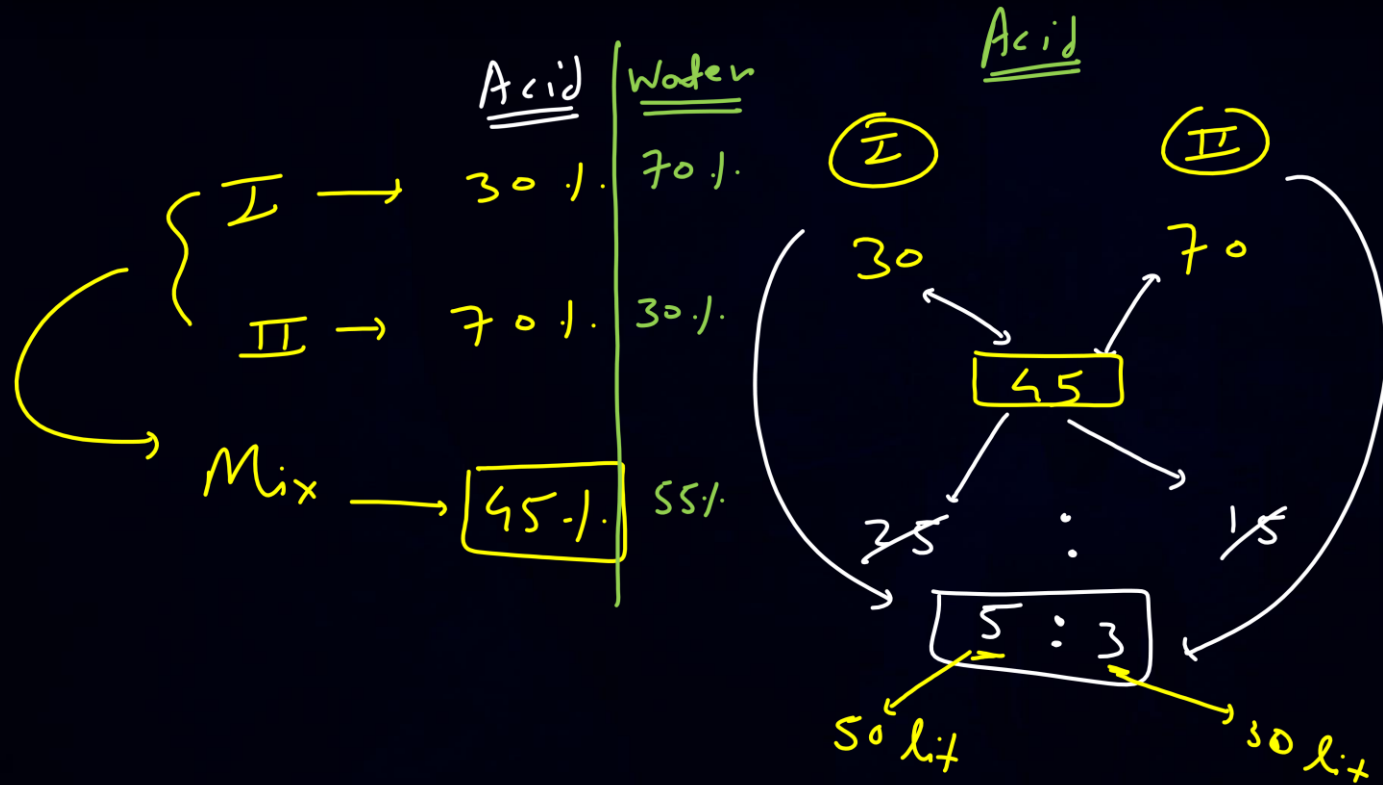
$$P\% = \frac{30}{120} \times 100 = \frac{300}{12} = 25\%$$



Mixture & Allegation



Mixtures & Allegations :





Mixtures & Allegations :



① Factor should be same

②



$$P_1 < P_{mix} < P_2$$

or

$$P_2 < P_{mix} < P_1$$



Example I: In what ratio must a grocer mix two varieties of pulses costing Rs. 15 and Rs. 20 per kg respectively so as to get a mixture worth Rs. 16.50 kg? Also find

$\text{Total} = 400 \text{ kg}$
 $\text{I} \quad \text{II}$
 $15 \quad 20$
 16.50
 $3.5 \quad 1.5$
 $25 : 15$
 $7 : 3 = 10$
 $I = \frac{7}{10} \times 400 = 280$

$$\begin{array}{l|l} \text{kg} & \text{I} \rightarrow 2 \\ & \text{II} \rightarrow 48 \end{array}$$

$$\frac{\text{I}}{\text{II}} = \frac{7}{3}$$

$$\frac{y}{\cancel{18}^{\cancel{148}}_{16}} = \frac{7}{8},$$

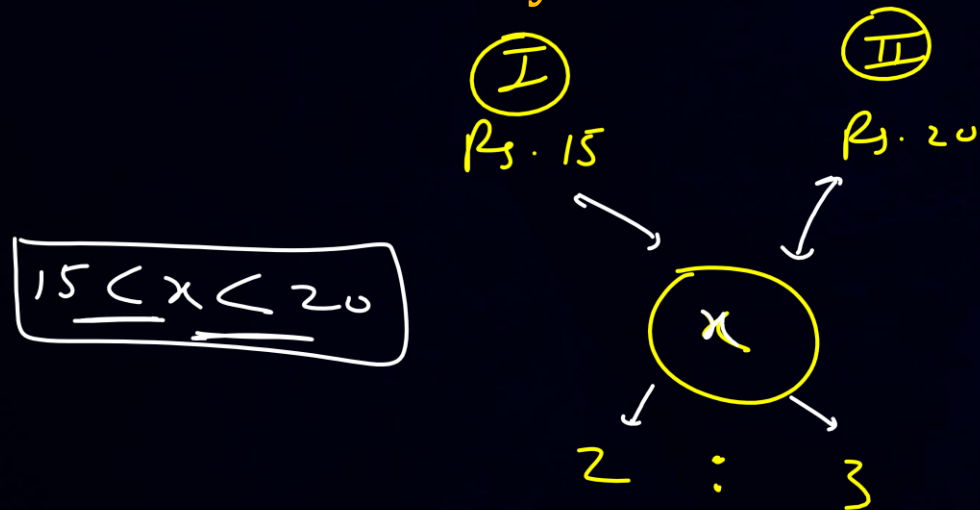
$$\boxed{y = 112}$$



Mixtures & Allegations :



Example II: The cost of Type 1 rice is Rs. 15 per kg and Type 2 rice is Rs. 20 per kg. If both Type 1 and Type 2 are mixed in the ratio of 2 : 3, then the price per kg of the mixed variety of rice is:



$$\frac{I}{II} = \frac{2}{3}$$
$$\frac{(20 - x)}{(x - 15)} \times \frac{2}{3}$$
$$60 - 3x = 2x - 30$$
$$90 = 5x$$
$$18 = x$$

QUESTION- 2



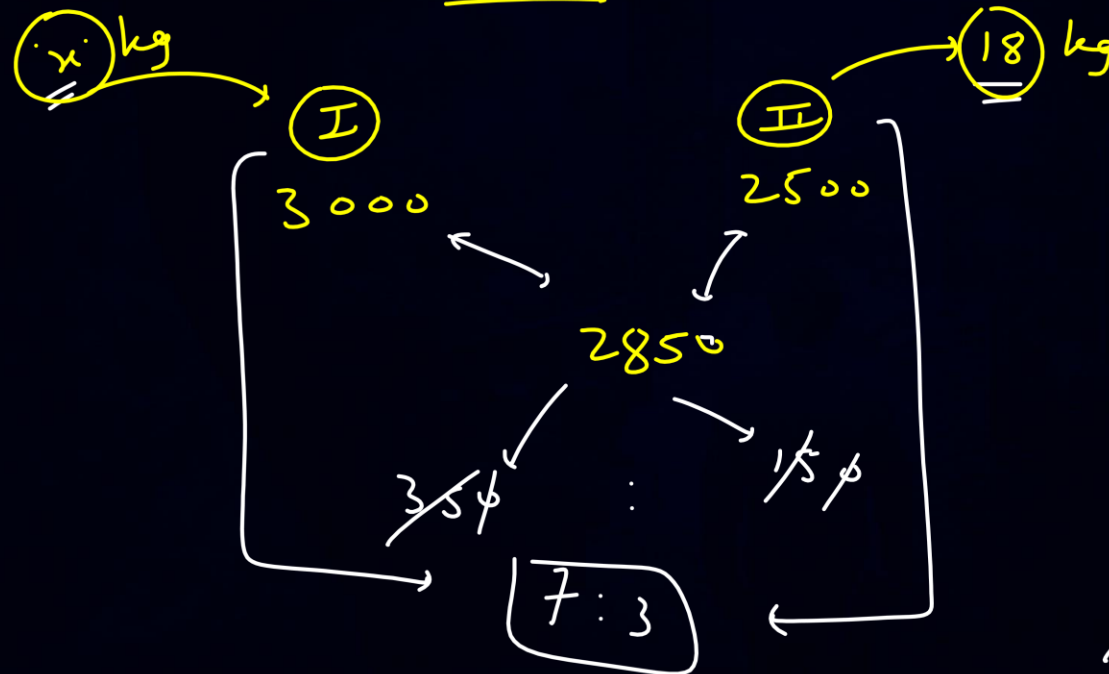
#Q. Rajni bought x kg of coffee that cost INR 3000 per kg and 18 kg of coffee at INR 2500 per kg for the company picnic. Find the total number of kilogram of coffee purchased, if the average cost per kg of both types together is INR 2850.

A 42 kg

B 18 kg

C 63 kg

D 60 kg



$$\begin{aligned} \text{Total} &= x + 18 \\ &= 42 + 18 \\ &= \underline{\underline{60 \text{ kg}}} \end{aligned}$$

$$\frac{II}{I} = \frac{7}{3}$$

$$\frac{x}{18} \times \frac{7}{3}$$

$$x = 18 \times \frac{7}{3}$$

$$x = 42 \text{ kg}$$

QUESTION- 3



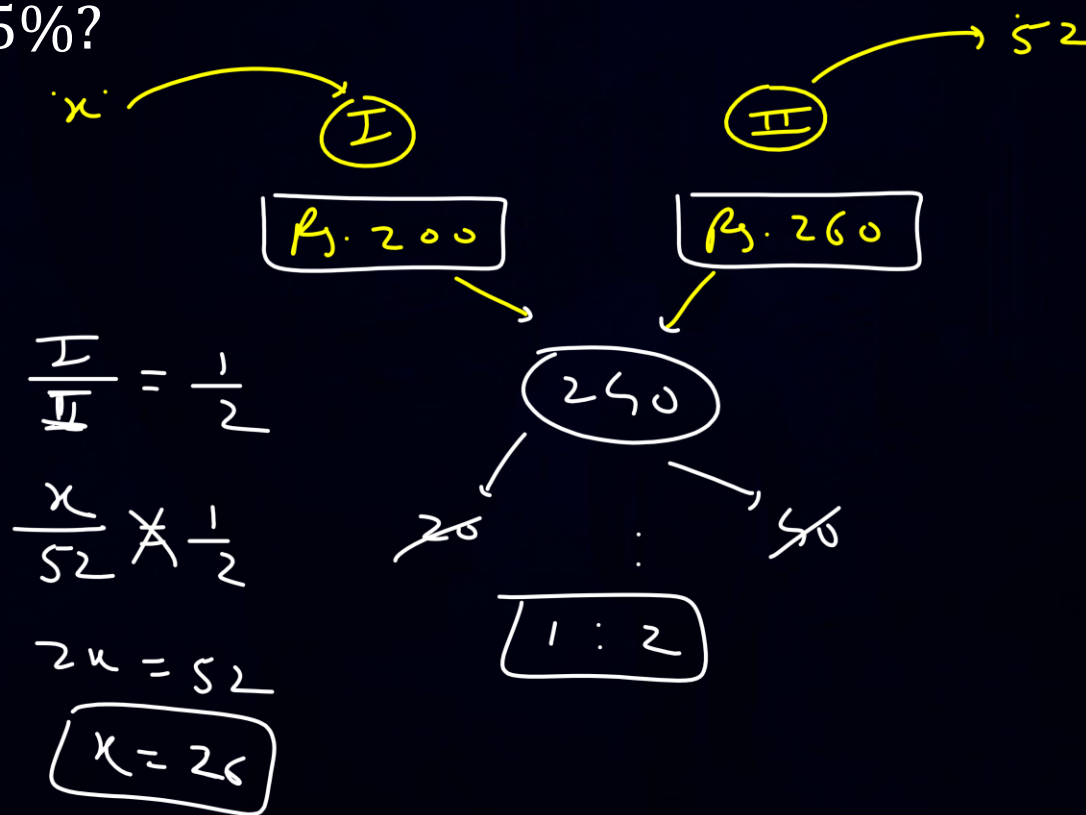
#Q. A merchant purchased two qualities of pulses at the rate of Rs. 200 per quintal and Rs. 260 per quintal. In 52 quintals of the second quality, How much pulse of the first quality should be mixed so that by selling the resulting mixture at Rs.300 per quintal, he gains a profit of 25%?

A 100 quintals

B 104 quintals

C 26 quintals

D None of these



$$\begin{aligned}
 &SP_{mix} = Rs. 300 \\
 &P = 25\% \\
 &SP = \frac{(100 + P)}{100} \cdot CP \\
 &\frac{300}{100} = \frac{125}{100} \cdot CP \\
 &240 = CP_{mix}
 \end{aligned}$$

QUESTION- 4

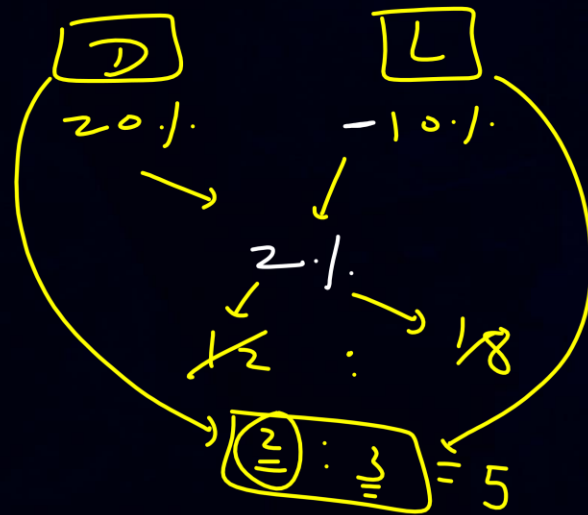
Loss % $\rightarrow$ - Profit %



#Q. A person spent Rs 50000 to purchase a desktop computer and a laptop computer. He sold the desktop at 20% profit and the laptop at 10% loss. If overall he made a 2% profit then the purchase price, in rupees, of the desktop is [CAT 2020 : Slot 1]

$$\text{T. Spending} = \text{Rs. } 50000$$

$$\begin{aligned} 2 - (-10) \\ = 2 + 10 \\ = 12 \end{aligned}$$



$$D = \frac{2}{5} \times \frac{10000}{50000} = \underline{\underline{20000}}$$

QUESTION- 5

1 kg $\rightarrow$ 1000 gm



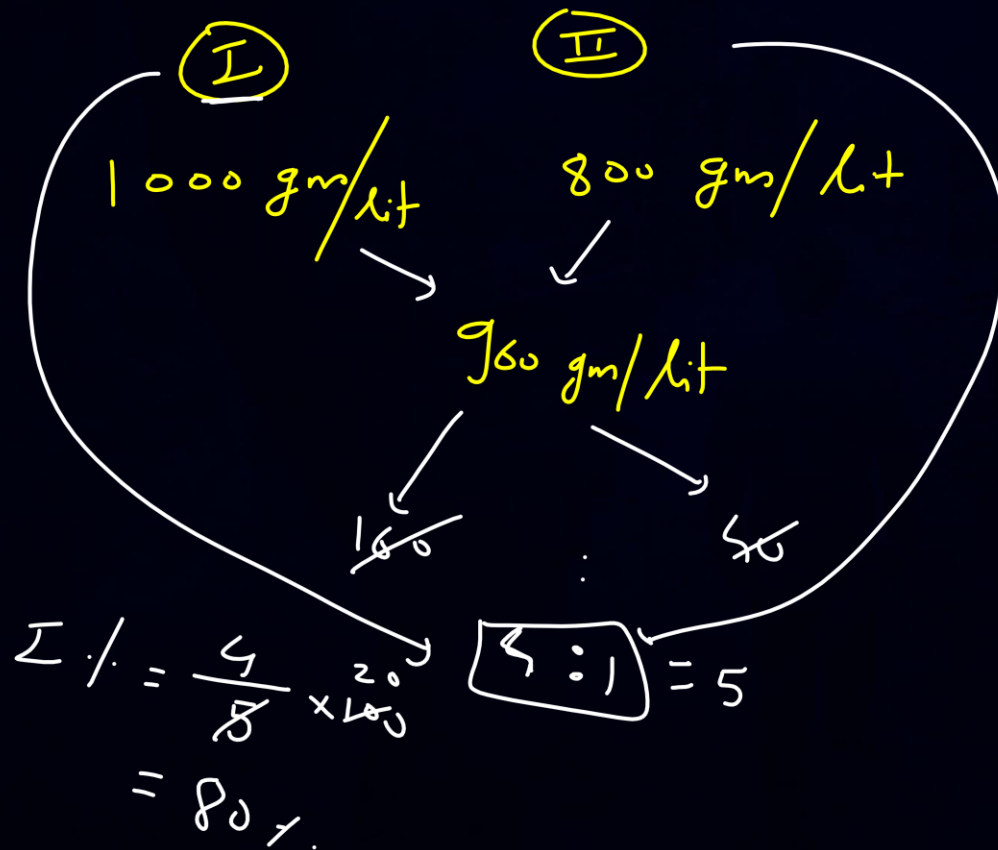
#Q. A chemist mixes two liquids 1 and 2. One litre of liquid 1 weighs 1 kg and one litre of liquid 2 weighs 800 gm. If half litre of the mixture weighs 480 gm, then the percentage of liquid 1 in the mixture, in terms of volume, is [CAT 2019 : Slot 1]

A 70

B 85

C 80

D 75



$$\begin{aligned} \text{Mix} &\rightarrow 480 \text{ gm} / \frac{1}{2} \text{ lit} \\ &\rightarrow 960 \text{ gm/lit} \checkmark \end{aligned}$$

QUESTION- 6



#Q. Bottle 1 contains a mixture of milk and water in 7 : 2 ratio and Bottle 2 contains a mixture of milk and water in 9 : 4 ratio. In what ratio of volumes should the liquids in Bottle 1 and Bottle 2 be combined to obtain a mixture of milk and water in 3 : 1 ratio?

[CAT 2017 : Slot 2]

$$\frac{M}{W} = \frac{3}{1}$$

$$\frac{7x + 9y}{2x + 4y} = \frac{3}{1}$$

$$7x + 9y = 6x + 12y$$

$$x = 3y$$

A 27 : 14 *let.*

B 27 : 13 *let.*

$$\frac{x}{y} = \frac{3}{1}$$

$$x = 3, y = 1$$

C 27 : 16

D 27 : 18

I = $9x$

M : W
7 : 2

M = $7x$
W = $2x$

II = $13y$

M : W
9 : 4

M = $9y$
W = $4y$

M = $7x + 9y$
W = $2x + 4y$

$$\frac{M}{W} = \frac{3}{1}$$

$$\frac{\frac{I}{II}}{\frac{I}{II}} = \frac{9x}{13y}$$

$$\frac{I}{II} = \frac{9(3)}{13(1)} = \frac{27}{13}$$

M : W

I	7 : 2 = 9
II	9 : 4 = 13
Mix	3 : 1 = 4

I $\frac{2}{9} (9 \times 4 \times 13)$

II $\frac{5}{13} (9 \times 4 \times 13)$

$\frac{1}{4} (9 \times 4 \times 13)$

I 104

II 144

232 : 13

QUESTION- 7

$$I : II : III = 1 : 2 : 3$$



#Q. Three vessels having volumes in the ratio of 1:2:3 are full of a mixture of coke and soda. In the first vessel, ratio of coke and soda is 2:3, in second 3:7 and in third 1:4. If the liquid in all the three vessels were mixed in a bigger container, what is the resulting ratio of coke and soda?

A ~~4:11~~

B 5:7

C 7:11

D 7:5

$$\begin{aligned} I &= 10 \\ C &= \frac{2}{5} \times 10 = 4 \\ S &= 10 - 4 = 6 \end{aligned}$$

$$\begin{aligned} C:S \\ 2:3 = 5 \end{aligned}$$

$$\begin{aligned} II &= 20 \\ C &= \frac{3}{10} \times 20 = 6 \\ S &= 20 - 6 = 14 \end{aligned}$$

$$\begin{aligned} C:S \\ 3:7 = 10 \end{aligned}$$

$$\begin{aligned} III &= 30 \\ C &= \frac{1}{5} \times 30 = 6 \\ S &= 30 - 6 = 24 \end{aligned}$$

$$\begin{aligned} C:S \\ 1:4 = 5 \end{aligned}$$

$$\begin{aligned} C &= 4 + 6 + 6 = 16 \\ S &= 6 + 14 + 24 = 44 \\ C:S &= 16:44 = 4:11 \end{aligned}$$

QUESTION- 8

$$M : S = 1 : 3$$



#Q. A mixture contains lemon juice and sugar syrup in equal proportion. If a new mixture is created by adding this mixture and sugar syrup in the ratio 1 : 3, then the ratio of lemon juice and sugar syrup in the new mixture is [CAT 2022 : Slot 1]

$$\begin{array}{l} \text{Mix} = 100 \\ L = 50 \\ S = 50 \\ L : S \\ 1 : 1 \end{array} \quad \begin{array}{l} \text{Sugar} = 300 \\ L = 0 \\ S = 300 \end{array}$$

A 1 : 6

B 1 : 4

C 1 : 5

D 1 : 7

$$\begin{array}{l} \text{New} \\ L = 50 + 0 = 50 \\ S = 50 + 300 = 350 \\ L : S = 50 : 350 \\ = 1 : 7 \end{array}$$

QUESTION- 9

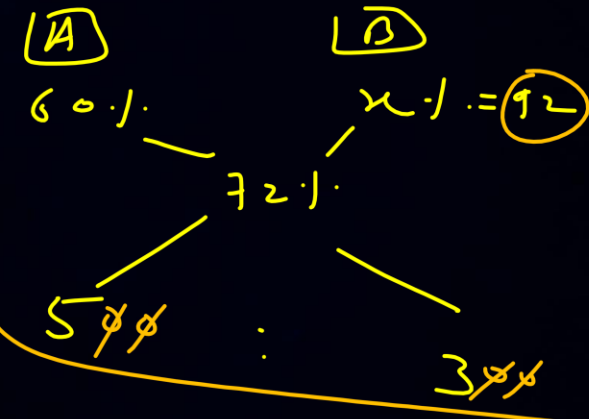


#Q. Two alcohol solutions, A and B, are mixed in the proportion 1:3 by volume. The volume of the mixture is then doubled by adding solution A such that the resulting mixture has 72% alcohol. If solution A has 60% alcohol, then the percentage of alcohol in solution B is

$$60 < 72 < x$$

$$A : B = 1 : 3$$

~~205~~
123 [CAT 2020 : Slot 3]



$$\begin{array}{l} A = 100 \\ B = 300 \\ \hline 400 \end{array}$$

$$\begin{array}{l} \text{Adj} \\ (\text{Sol}^n \text{ A}) \\ + 400 \\ \hline \end{array}$$

$$\begin{array}{l} A = 500 \\ B = 300 \\ \hline 800 \end{array}$$

A 94%

B ~~92%~~

C 90%

D 89%

$$\begin{aligned} 60\% \text{ of } A + x\% \text{ of } B &= 72\% \text{ of } 800 \\ \left(\frac{60}{100} \times 500 \right) + \left(\frac{x}{100} \times 300 \right) &= \frac{72}{100} \times 800 \end{aligned}$$

$$\begin{aligned} 300 + 3x &= 576 \\ 3x &= 276 \\ x &= 92 \end{aligned}$$

QUESTION- 10



#Q. A vessel contained a certain amount of a solution of acid and water. When 2 litres of water was added to it, the new solution had 50% acid concentration. When 15 litres of acid was further added to this new solution, the final solution had 80% acid concentration. The ratio of water and acid in the original solution was

A 5 : 3

B 4 : 5

C 5 : 4

D ~~3 : 5~~

[CAT 2024 : Slot 2]

original

Acid = x
 water = y

W : A
 $y : x$
 $3 : 5$

⊕ 2 lit

I (New Solⁿ)

Acid = x
 water = $z + y$

A : W
 Solⁿ : 50%
 $1 : 1$
 $\frac{x}{z+y} = \frac{1}{1}$
 $x = z + y$

⊕ 15 lit

II Final

Acid = $x + 15$
 water = $y + z$

A : W
 Solⁿ : 80%
 $4 : 1$
 $\frac{x+15}{y+z} = \frac{4}{1}$

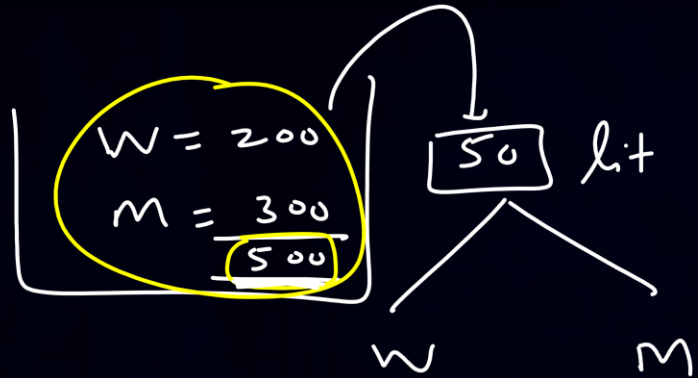
$x = z + y$ → $z + y + 15 = 4y + 8$
 $\Rightarrow y + 15 = 4y + 8$

$\therefore x = z + 3 = 5$

$y = 3y$
 $3 = y$



Replacement & Removal :



$$= \frac{1}{10} \times 200 = \frac{1}{10} \times 300$$

$$= (20) = (30)$$

Removed

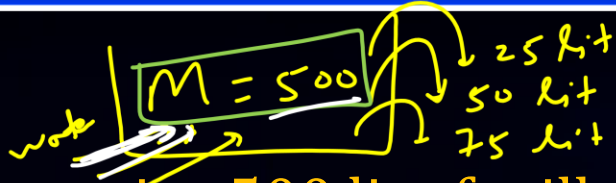
$$= \frac{50}{500} = \frac{1}{10}$$



Replacement & Removal :



Example:



A vessel contains 500 lit of milk, 25 lit of milk taken out from it and replaced by water. Then again from mixture 50 lit are taken out and replaced by water. Then again from mixture 75 lit are taken out and replaced by water. Find

- 1) Quantity of milk in the resultant mixture
- 2) Quantity of water in the resultant mixture
- 3) Ratio of Milk and Water in the resultant mixture

$$\textcircled{1} \cancel{500} \times \frac{19}{20} \times \frac{9}{10} \times \frac{17}{20} = \text{New Milk}$$

$$\frac{171 \times 17}{8} = \text{N. Milk}$$

$$\frac{2907}{8} = \text{N. Milk}$$

$$\boxed{363 \approx \text{N. Milk}}$$

$$\begin{aligned} \textcircled{II} \text{ Water} &= \text{Total} \\ &\quad - \text{N. Milk} \\ &= 500 - 363 \\ &= \boxed{137} \end{aligned}$$

$$\textcircled{III} \text{ M:W} = \boxed{363:137}$$

	<u>Removed</u>	<u>Remaining</u>
$\textcircled{I}$	$\frac{25}{500} = \frac{1}{20}$	$\frac{500-25}{500} = \frac{475}{500}$
$\textcircled{II}$	$\frac{50}{500} = \frac{1}{10}$	$\frac{475-50}{475} = \frac{425}{475}$
$\textcircled{III}$	$\frac{75}{500} = \frac{3}{20}$	$\frac{425-75}{425} = \frac{350}{425}$



Replacement & Removal :



Example:

$$\begin{array}{l} m = 200 \\ w = 300 \\ \hline 500 \end{array}$$

25 lit
50 lit

A vessel contains mixture of 200 lit milk and 300 lit water. 25 lit of mixture is taken out from it and replaced by water. Then again from mixture 50 lit are taken out and replaced by water. Find

- 1) Quantity of milk in the resultant mixture
- 2) Quantity of water in the resultant mixture
- 3) Ratio of Milk and Water in the resultant mixture

$$\textcircled{1} \text{ 25 lit } \times \frac{19}{20} \times \frac{9}{10} = \text{N. Milk}$$

$$\boxed{171 = \text{N. Milk}}$$

$$\textcircled{2} \text{ Water} = \text{Total} - \text{N. Milk} \\ = 500 - 171 = 329$$

$$\textcircled{3} m : w = 171 : 329$$

	<u>Removed</u>	<u>Remaining</u>
$\textcircled{I}$	$\frac{25}{500} = \frac{1}{20}$	$\frac{20-1}{20} = \left(\frac{19}{20}\right)$
$\textcircled{II}$	$\frac{50}{500} = \frac{1}{10}$	$\frac{10-1}{10} = \left(\frac{9}{10}\right)$

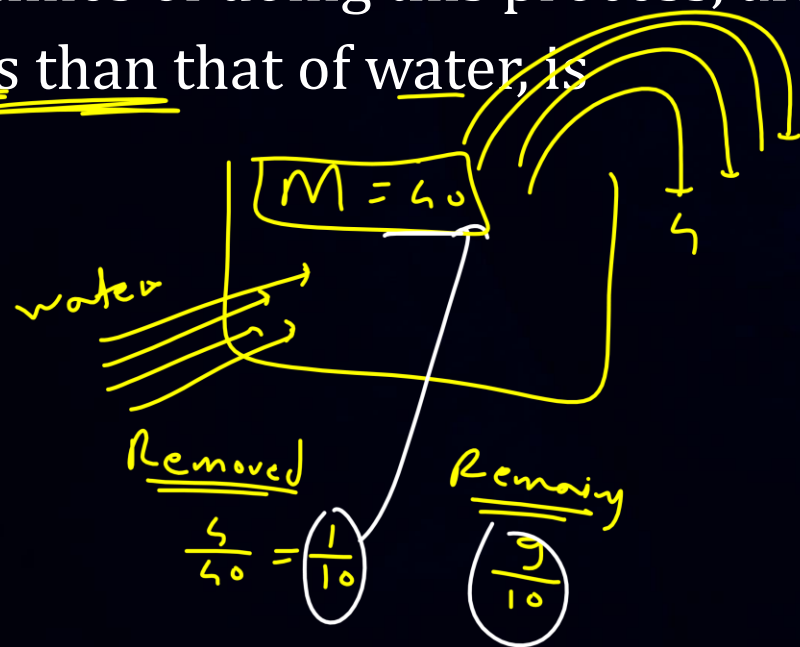
QUESTION- 11

7



#Q. A container has 40 liters of milk. Then, 4 liters are removed from the container and replaced with 4 liters of water. This process of replacing 4 liters of the liquid in the container with an equal volume of water is continued repeatedly. The smallest number of times of doing this process, after which the volume of milk in the container becomes less than that of water, is

[CAT 2023 : Slot 2]



$$M < w$$

New

$$M < 20$$

$$N \cdot \text{milk} < 20$$

$$40(0.9)^7 = 19.13 \dots$$

$$40 \times \left(\frac{9}{10}\right) \times \left(\frac{9}{10}\right) \times \left(\frac{9}{10}\right) \times \dots = N \cdot \text{milk}$$

$$40 \times \left(\frac{9}{10}\right)^n < 20$$

$$40(0.9)^n < 20$$

$$40(0.9)^7 < 20$$

$$19.13 < 20$$

QUESTION- 12



#Q. A glass is filled with milk. Two-thirds of its content is poured out and replaced with water. If this process of pouring out two-thirds the content and replacing with water is repeated three more times, then the final ratio of milk to water in the glass, is

TRY
[Next Class]

[CAT 2024 : Slot 1]

A 1 : 27

B 1 : 81

C 1 : 26

D 1 : 80



2 Mins Summary

- 1) Concept & Numerical based on 'Mixture & Allegation'
- 2) Concept & Numerical based on 'Replacement & Removal'
- 3) Problems for Practice & CAT PYQs



*Thank
You*